

Given $N \times N$ Weight matrix, W performs N^2 MACs and loads N^2 FP32 weights from memory

TASKS:

1. Given $N=512$

a. dense MVM Compute (FLOPs)

2 FLOPs each one for multiplier and the other for accumulator.

for $N \times N$ matrix

$$\Rightarrow 512 \times 512$$

$$\Rightarrow 262,144 \text{ elements in the matrix}$$

FP32 weight 4 Bytes each

INT32 weight 4 Bytes each

for dense MVM Compute

$$= 2 \times N^2$$

$$= 2 \times 262,144$$

$$= 524,288 \text{ FLOPs}$$

(2 FLOPs = one for multiplier
one for accumulator)

b. Dense Memory Bytes

$$= 4 \times N^2$$

$$= 4 \times 262,144$$

$$= 1,048,576 \text{ Bytes}$$

(FP32 4 bytes each)

c7. Sparse Compute (FLOPs, as a function of s)

Only non-zero weights participate in MACs.

With sparsity s , the fraction of non-zeros is $(1-s)$.

$$\begin{aligned}\text{Sparse FLOPs} &= 2 \times (1-s) \times N^2 \\ &= 2 \times (1-s) \times 262,144 \\ &= \underline{524,288 (1-s) \text{ FLOPs}}\end{aligned}$$

d7. Sparse Memory Bytes (as function of s)

CSR format uses three arrays:

i. values array = 4 Bytes $\times$ NNZ = $4N^2(1-s)$

ii. Column indices = 4 Bytes $\times$ NNZ = $4N^2(1-s)$

iii. Row pointers = 4 Bytes $\times$ (N+1) = $4(N+1)$

$$\begin{aligned}\text{So, sparse Bytes} &= 4N^2(1-s) + 4N^2(1-s) + 4(N+1) \\ &= 8N^2(1-s) + 4(N+1)\end{aligned}$$

$$\begin{aligned}(N=512) \quad &= 8 \times (262,144)(1-s) + 4(512+1) \\ &= \underline{2,097,152 (1-s) + 2,052 \text{ Bytes}}\end{aligned}$$

2. Theoretical speedup of sparse vs Dense MVM (FLOPs)

$$\text{Speedup} = \frac{\text{Dense FLOPs}}{\text{Sparse FLOPs}}$$

$$= \frac{2N^2}{2(1-s)N^2}$$

$$= \frac{1}{(1-s)}$$

For 2x Speedup,

$$2 = \frac{1}{(1-s)}$$

$$1-s = \frac{1}{2}$$

$$\Rightarrow 2 - 2s = 1$$

$$\Rightarrow 2s = 1$$

$$\Rightarrow s = 0.5 \Rightarrow \text{50 FLOPs speedup becomes 2x at 50\% sp}$$

So, at 50% Sparsity, the sparse format cuts the number of MACs exactly in half.

3. Memory Breakeven sparsity

Sparse Bytes = Dense Bytes given

$$8(1-s)N^2 + 4(N+1) = 4N^2$$

$$N = 512,$$

$$\Rightarrow 8(1-s)262,144 + 4(513) = 4 \times 262,144$$

$$\Rightarrow 2,097,152(1-s) + 2,052 = 1,048,576$$

$$\Rightarrow 2,097,152(1-s) = 1,046,524$$

$$\Rightarrow (1-s) = 1,046,524 / 2,097,152$$

$$\Rightarrow (1-s) = 0.4990215301813$$

$$\Rightarrow s = 0.500978$$

$$\Rightarrow s = 0.501$$

Memory Breakeven occurs at ~~above~~ 50.01% ~~s~~ above
50.1% sparsity, CSR uses less memory than dense storage.

4. End-to-End speedup at $s=0.9$ for memory bound case
 $s=0.9$

$$\text{Bandwidth} = 320 \text{ GB/s}$$

from $\{b\}$ dense memory bytes = 1,048,576 Bytes

$$\text{Sparse}_{\text{memory}} \text{ Bytes} = 2,097,152(1-s) + 2,052 \text{ from } \{d\}.$$

$$\Rightarrow 2,097,152(1-0.9) + 2,052$$

$$\Rightarrow 2,097,152(0.1) + 2,052$$

$$\Rightarrow 211,767.2 \text{ Bytes}$$

For a memory-Bandwidth limited system

$$\text{Speedup} = \frac{\text{Dense Bytes}}{\text{Sparse Bytes}}$$

$$= \frac{1,048,576}{211,767.2}$$

$$= 4.9515$$

$$\Rightarrow \underline{\underline{4.95 \text{ times}}}$$